MILITARY SPECIFICATION

LENS, OPHTHALMIC, PLASTIC (HARD RESIN)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers ophthalmic lenses molded from a thermosetting resin, base monomer of allyl diglycol carbonate.

1.2 Classification. Lenses shall be of one type, simple vision, and of the following classes and styles, as specified (see 6.2).

Class.
- A. Simple
- B. Compound

Style.
1. Spheres.
2. Minus plano cylinders.
3. Plus on minus compounds.
4. Minus on minus compounds.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-up, Paperboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI Z 80.1-1972 American National Standard Requirements for First Quality Prescription Ophthalmic Lenses

(Application for copies should be addressed to the American National Standards Institute, Inc. 1430 Broadway New York, N.Y. 10018).

AMERICAN SOCIETY FOR TESTING and MATERIALS (ASTM) Standards:

D-542-50	Index of Refraction of Transparent Organic Plastics.
D-792-66	Specific Gravity and Density of Plastics by Displacement.
D-1003-61	Haze and Luminous Transmittance of Transparent Plastics.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS.

3.1 Material. The lens shall be fabricated of molded thermosetting plastic resin polymerized from allyl diglycol carbonate monomer and a catalyst with necessary additives to produce satisfactory ophthalmic lenses. The lens material shall conform to the physical requirements listed in table I when tested as specified in section 4.

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Table I. Physical Requirements

Characteristic	Requirement
Index of refraction, 20°C, n_D	1.498 ± 0.002
Specific gravity, 25°C	1.32 ± 0.02
Barcol hardness, minimum	26 (15 seconds)
Light transmission, 1/4-inch, per cent, minimum	89

The lens material shall have a high degree of clarity and transparency. The lenses shall remain clear and free of discoloration for a period of one (1) year minimum when stored at normal room temperature away from direct sunlight. Discoloration shall be determined by loss in light transmission. It shall be durable and of suitable composition and hardness to give good service in use, resist the action of weather, and not be easily scratched or damaged under ordinary conditions. It shall be free of bubbles, inclusions, striae and cloudiness.

3.2 Construction. The lens shall be molded from the material specified in 3.1, free from bubbles or defects which might result from the molding process. The lens shall be a member of an integrated series of ophthalmic lenses having surface curvatures computed to conform with the corrected curve concept. Cylinder and spherical cylinder lenses shall be fabricated in the minus cylinder form (cylinders on the concave side).

3.2.1 Refractive power. The refractive power shall correspond with the dioptric power specified by industry standards. Tolerances for refractive power shall be as follows:

0 through 9.00 diopter - ± 0.09 diopter
9.00 diopter and above - ± 0.12 diopter. In addition to the tolerances for axial and marginal power (3.2.5.4).

3.2.2 Thickness. Center thickness shall be in accordance with accepted commercial standards for ophthalmic lenses (dress eyewear) except that the center thickness shall be a minimum of 1.6 mm. The thickness at 1 1/2 mm from the peripheral edge of the lens shall be a minimum of 1 mm.

3.2.3 Size. The lens diameter shall not be less than 58 mm, except for plus spheres above 5.00 and spherical cylinders with sphere curves above 5.00 and cylinders up to 2.00. Any cloudiness, haze or peripheral defect shall not reduce the lens diameter significantly.

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3.2.4 Class A. Simple. The Class A lenses shall be plus spheres and minus spheres.

3.2.4.1 Style 1. Spheres. The single vision, uncut, simple lens shall be a plus sphere or minus sphere. The axial power shall correspond with plus or minus spherical dioptric power specified, with allowable tolerances. The marginal power shall correspond with axial spherical and cylindrical power as specified, with allowable tolerances.

3.2.5 Class B. Compound. The Class B lenses shall consist of plano cylinders and spherical cylinders.

3.2.5.1 Style 2. Minus plano cylinders. The minus plano cylinder lens shall have the cylinder curve molded on the concave side. Axial power shall correspond with minus cylindrical power specified and marginal power shall correspond with specified spherical and cylindrical power. Tolerances shall be in conformance with 3.2.5.4.

3.2.5.2 Style 3. Plus on minus compounds. The plus on minus lens shall have the cylinder curve on the concave side. Axial power shall correspond with plus spherical and minus cylindrical power as specified, with allowable tolerance (3.2.5.4). The marginal power shall correspond with axial and cylindrical power as specified, with allowable tolerance (3.2.5.4).

3.2.5.3 Style 4. Minus on minus compounds. The minus on minus lens shall correspond to minus spherical and minus cylindrical power as specified. The marginal power shall correspond with axial spherical and cylindrical power as specified. Tolerances shall be within the limits of 3.2.5.4.

3.2.5.4 Tolerances.

3.2.5.4.1 Axial cylindrical power. The refractive power measured through the optical center shall agree to the specified power, to within the following tolerances:

Powers zero through 6.00	± 0.06 diopters.
Powers above 6.00	± 1 percent of power.

3.2.5.4.2 Marginal power. The tolerances for marginal power shall be as follows:

(1) The marginal meridional power referred to the vertex sphere at points A and B (see figure 3 of standard referenced below) shall agree to the specified power within the tolerances for meridional power shown in tables 2 through 6, inclusive, of American National Standard Z80.1-1972.

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(2) The cylindrical power, referred to the vertex sphere at points A and B (see figure 3 of referenced standard) shall agree to the specified power within the tolerances for cylindrical power shown in tables 2 through 6 of ANSI Z80.1-1972. The marginal power shall be measured as specified in 4.4.3.2.

3.3 Impact test. The lens shall neither fracture nor chip when tested as specified in 4.4.1.4.

3.4 Workmanship. Workmanship shall be first class. The lens shall be colorless, smooth and free from blemishes (internal and external), striae, bubbles, chips, waves and other surface and internal defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the lenses, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the name of the component/material and test results.

4.2 Sampling.

4.2.1 Sampling for tests and examination. Sampling for tests and examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table II. Unit of product for examination and test purposes shall be one lens.

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Table II. Sampling plans.

For test	Inspection level	AQL
Refractive index and specific gravity	S-2	*
Barcol hardness and light transmission	S-1	*
Impact test	S-3	*
For examination		**Major**
Power, thickness, yellowing and distortion.	S-3	1.0
Size, clarity, striae, blemishes, bubbles, etc.	II	Minor 2.5

* Acceptance number shall be zero.

4.3 Examination. The lens shall be examined for defects including, but not limited to, those listed in table III.

Table III-Classification of defects.

Major	Defect
101	Refractive power not in tolerance.
102	Lens distorted*.
103	Cylinders not in minus cylinder form.
104	Yellowing*.
105	Lens below minimum thickness
	*visible to naked eye.
Minor	
201	Waves, striae, bubbles, chips, blemishes, etc.
202	Size not as specified.
203	Cloudiness or haze.

4.4 Tests. The lens samples shall be tested for compliance with requirements of Section 3.

4.4.1 Test methods. Test methods shall be as specified in Table IV.

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Table IV - Methods

Characteristic	Requirement paragraph	Test Method
Index of refraction	3.1	ASTM D-542-50
Specific gravity	3.1	ASTM D-792-66
Barcol hardness	3.1	par. 4.4.1.1
Light transmission	3.1	ASTM D-1003-61
Refractive powers	3.2	par. 4.4.1.2
Axial and Marginal powers	3.2.5.4	par. 4.4.1.3
Impact test	3.3	par. 4.4.1.4
Defects	3.4	par. 4.4.1.5

4.4.1.1 Barcol hardness. The barcol hardness shall be conducted as follows:

4.4.1.1.1 Apparatus. The apparatus shall consist of the following:

(a) A Barcol Impressor manufactured by the Barber-Colman Company, Rockford, Illinois.

(b) A pressure applicating device, similar to the one shown in the accompanying diagram (Figure 1). This is comprised essentially of the following:

(1) Two 1/2" x 2" springs, the upper one applying the pressure to the test specimen and the lower one acting as a shock absorber. These should be so adjusted by the locking nuts that the upper spring is exerting a downward force of approximately 35 lbs. and the lower spring an upward force of approximately 15 lbs., giving a resultant force of 20 lbs. acting downward on the test specimen.

(2) A vertical rod working freely through two sleeves. The Barcol Impressor is rigidly attached to the bottom of this rod. The rod is raised by means of the lever and is forced downward by the 20 lb. spring action.

(3) An adjustable table attached to the base of the apparatus. The plastic test specimen is placed on this table and is moved (either upward or downward) by means of the screw adjustment until the 20 lb. downward force of the Barcol needle is applied to the plastic sample. When initially setting up the apparatus, it is advisable to move the plastic test specimen to the proper position and to scribe a reference line on the moveable vertical rod at a point even with the top surface of the upper sleeve. The rod should then be raised and a second scribe line made on the rod at a point 3/32 in. below the first line. In testing specimens of varying thickness it is then only necessary to adjust the table until the upper scribe line is even with the top surface of the upper sleeve. In applying the load for the actual Barcol readings, the rod is raised until the lower scribe line is even with the upper surface of the sleeve. The rod is then quickly released.

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4.4.1.1.2 Test specimens. Test specimens shall be lenses or flat pieces molded from lens batch. Lens should be measured on plano or nearly flat area, thickness shall be 3.0 mm or more. Flat piece shall be approximately 1 inch by 3 inches by 1/8 inch minimum thickness.

4.4.1.1.3 Conditioning. Specimens shall be conditioned for at least 24 hours at 23° ± 1.1°C. (73.4° ± 2°F and 50 ± 2 per cent relative humidity prior to testing. The actual tests shall be conducted at the same atmospheric conditions.

4.4.1.1.4 Procedure

(a) The actual testing shall be carried out in an atmosphere maintained at 23° ± 1.1°C (73.4° ± 2°F) and 50 ± 2 per cent relative humidity.

(b) The specimen shall be placed on the adjustable table and moved into contact with the Barcol needle until the 20 lb. downward force is exerted by the spring system. (Move table until the upper reference mark on the adjustable vertical rod is even with the top surface of the upper sleeve of the vertical rod).

(c) Raise the adjustable rod 3/32 inch upwards by means of the lever (Raise until the lower of the two reference marks on the vertical rod is even with top surface of the upper sleeve).

(d) Move the test specimen slightly so that the Barcol needle will not contact a Barcol indentation mark or any surface irregularity. Release the lever quickly, but without jerking, and record the highest reading on the dial gage.

(e) Record the dial gage reading 15 seconds after applying the load.

(f) Repeat steps (d) and (e) until 5 separate hardness values have been obtained. Report average 15 second reading.

4.4.1.2 Refractive power. The power shall be measured with a lens measuring instrument.

4.4.1.3 Axial and marginal powers.

4.4.1.3.1 Axial powers. The axial powers may be measured on any standard lens measuring instrument capable of yielding the required accuracy.

4.4.1.3.2 Marginal powers. The marginal powers may be measured on any standard lens measuring instrument that has been appropriately modified to yield the required accuracy referred to the vertex sphere.

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One modification of a standard instrument that is satisfactory, is to add a hemispherical surface (see figure 3)[1] to act as the seating surface against which the lens under test is held. The radius of the surface is 27 mm. This surface is so arranged that the edge of the standard opening is tangent to the surface of the sphere. A dot is marked on the sphere to locate a point 30 degrees from the axis. The optical center of the lens is pressed against this dot during the test. Suitable prisms are used to bring the image of the tangent within the field of view of the observing telescope. When the measurements are made in this manner, the observed values are the values referred to the vertex sphere and no corrections are necessary. "[1] ANSI Z80.1"

4.4.1.4 Impact test. The impact test specified in 3.3 shall be conducted on sample units selected in accordance with specified sampling plan. The test shall be conducted as specified below: A 5/8-inch steel ball weighing approximately 0.56 ounce shall be dropped from a height of 50 inches upon the horizontal upper surface of the lens. The ball shall strike within a 5/8-inch diameter circle located at the geometric center of the lens. The ball shall be guided, but not restricted, in its fall by being dropped through a tube extending to within approximately 4 inches of the lens. In order to pass the test, the lens must not fracture (for the purpose of this test, a lens will be considered to have fractured if it cracks through its entire thickness and across a complete diameter into two or more separate pieces or if any lens material visible to the naked eye becomes detached from the ocular surface). The test shall be conducted with the lens supported by a tube (1-inch inside diameter, 1-1/4 inch outside diameter, and approximately 1-inch high) affixed to a rigid iron or steel base plate. The total weight of the base plate and its rigidly attached fixtures shall be not less than 27 pounds. The tube shall be made of rigid acrylic plastic, steel, or other suitable substance and shall have on the top edge a 1/8 - by 1/8-inch neoprene gasket having a hardness of 40 ± 5, as determined by ASTM Method D 1415; a minimum tensile strength of 1,200 pounds, as determined by ASTM Method D 412; and a minimum ultimate elongation of 400 percent, as determined by ASTM Method D 412. The diameter and/or contour of the lens support shall be modified as necessary so that the 1/8-by 1/8-inch neoprene gasket supports the lens at its periphery.

4.4.1.5 Defects. Lenses shall be examined for defects and other irregularities affecting the optical qualities, by moving the lens in front of the aperture of a lens measuring instrument. Sudden jumps or blurring of the image will indicate a defect.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. Two lenses (1 pair) of the stock number specified constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	10 units	360 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each lens shall be placed in a plastic envelope or shall be wrapped in tissue or other suitable material. The lens shall be further packaged in a paper envelope of appropriate size and design.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-676 except that commercial colors are acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Special requirements specified in Table II of PPP-B-636 shall apply for fiberboard boxes.

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5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard boxes shall be closed by method II of PPP-B-636. Special requirements of Table I of PPP-B-636 shall apply for PPP-B-636 boxes.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Individual lens envelope. In addition to any commercial markings, each envelope shall be marked with the power, name or registered trade mark of contractor, month and year of manufacture and the legend: "1/2 pair".

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended Use. The lenses covered by this specification are intended as first quality prescription ophthalmic lenses. Only single vision, uncut lenses are covered. The lenses conform to impact resistant requirements of FDA regulations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class, style, and power required (see 1.2).
(c) Selection of applicable levels of packing.

6.3 This specification does not include all types, classes, styles or sizes of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the items, which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6540-1024

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CATEGORY: N

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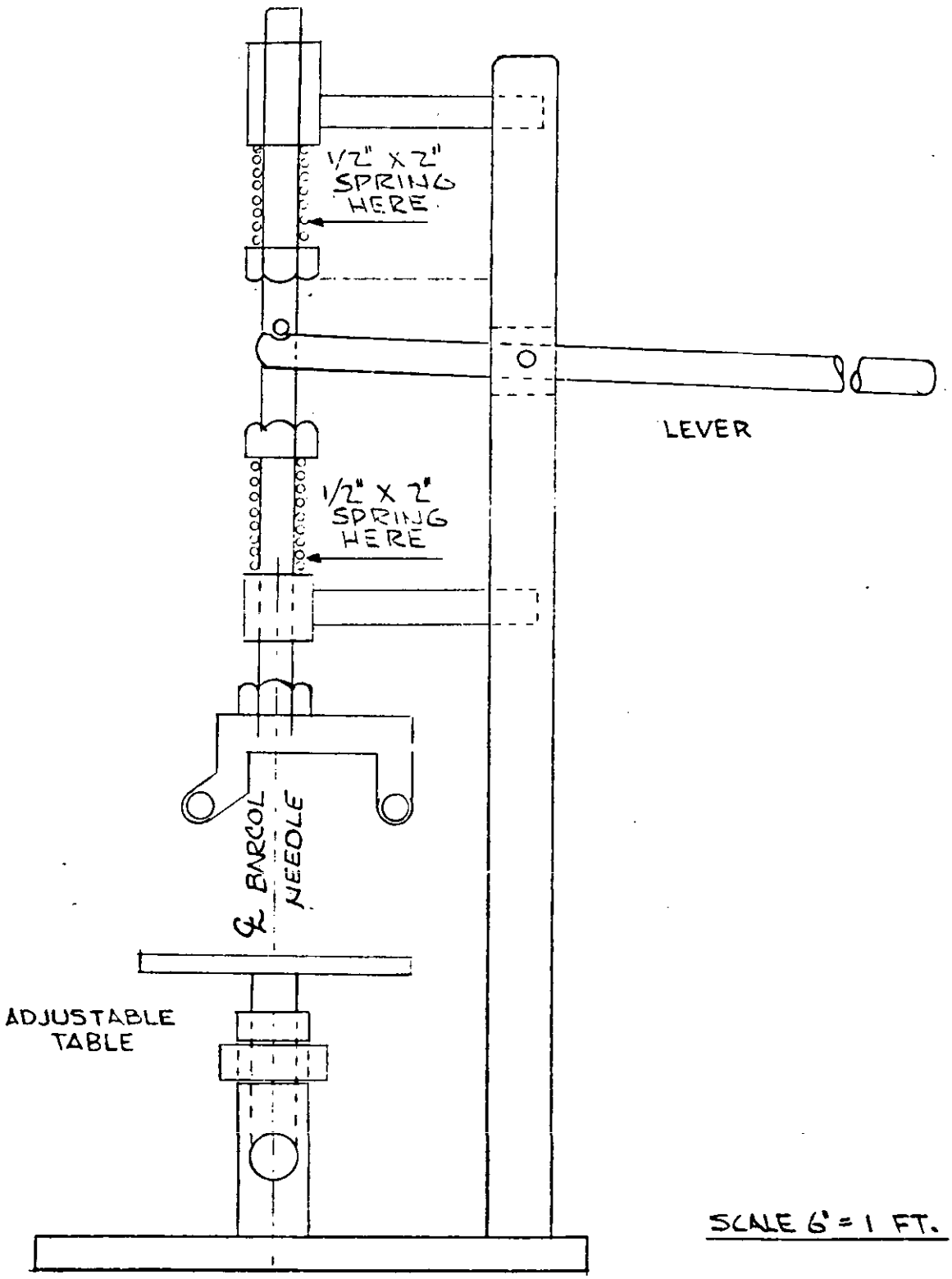


DIAGRAM OF BARCOL HOLDER

FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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